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BRIEF REPORT

# It's Time to Play: Peer-Mediated Intervention With a Student With Autism at Recess

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**Received:** 4 December 2024 | **Revised:** 4 December 2024 | **Accepted:** 14 February 2025

**Keywords:** autism spectrum disorder | inclusion | peer-mediated intervention | pivotal response training | play

## ABSTRACT

Students with autism spectrum disorder (ASD) are often taught within inclusive classrooms, which provide opportunities to develop social communication and peer interaction skills. Pivotal response training (PRT) and peer-mediated intervention (PMI) are evidence-based practices that promote social skill development. We used a concurrent multiple probe across peers design to evaluate the effects of training three third-grade peers to use PRT as a PMI to promote the social communication and play behaviors of a classmate with ASD during recess. Generalization measures were conducted in the classroom during free play. Results indicated a functional relationship between training peers on PRT and increased use of PRT and between peers' use of PRT and the target child's communication to peers, confirmed by Tau-U and visual analysis. A functional relation between peers' use of PRT and the target child's play with peers was not demonstrated. Similar effects were observed in the generalization environment.

## 1 | Introduction

A critical component of instruction for students diagnosed with autism spectrum disorder (ASD) is the promotion of social interaction skills. Many students with ASD are taught within inclusive classrooms that offer increased opportunities for learning and practicing social interaction and communication skills with peers. However, placement in a classroom with typically developing students without the use of systematic intervention is unlikely to increase the social interaction ability of students with ASD (Hume et al. 2019).

Peer-mediated intervention (PMI) is a systematic intervention approach that engages peers as the intervention agent to promote skill acquisition or generalization in target children. PMI involves training peers to provide increased opportunities for learning social interactions and other skills within natural environments (Neitzel 2008). Pivotal response training (PRT) is a naturalistic, evidence-based behavioral intervention to teach children with ASD social, communication, and play skills (Koegel and Frea 1993). PRT involves using choice and preferences, task interspersal, natural reinforcement, and procedures designed for ease of implementation by natural change agents (Koegel and Koegel 2019). A promising approach for promoting the acquisition of social and play skills for students with ASD in the natural environment is to combine PMI and PRT by using peers as natural change agents to implement PRT (Pierce and Schreibman 1995, 1997).

Combining PRT and PMI to promote social and play skill development of elementary-aged students with ASD was first evaluated by Pierce and Schreibman (1995, 1997). Their research demonstrated that peers could be trained to use PRT with children with ASD, but the initial study lacked measures to assess generalization (Pierce and Schreibman 1995), and a shortcoming of these early studies may be the application of the intervention in a contrived setting of an empty classroom (Pierce and Schreibman 1995, 1997). For PMI to be meaningful

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in the school environment, assessments of the generalization of learned skills need to occur outside of the training location in novel social situations (Chang and Locke 2016).

The purpose of the present study was to examine the effectiveness of training peers in the natural recess environment to deliver PRT with a child with ASD and the generalization of their use of the intervention in an untrained, natural, free-play environment. An additional purpose of the study was to examine the effects of peer-implemented PRT on the social communication and play behavior of a child with ASD. Specifically, the study addressed the following research questions:

1. When trained on using PRT with a child with ASD, will third-grade peers learn the skills, apply them during recess, and generalize skills to an untrained free-play environment?
2. Is there a relationship between peers' use of PRT strategies with a child with ASD and the child's social communication and play behaviors?

## 2 | Method

### 2.1 | Participants

Eve, the target child, was 8 years old. Eve is a White female with ASD. She was diagnosed with ASD at age 4 by a developmental pediatrician not associated with this study. Eve was enrolled at a private Waldorf school. She often spoke in five- to seven-word sentences and frequently used language to request items and activities. Eve's teacher and family expressed concerns that she was not interacting with her peers during recess. Eve's teacher nominated three typically developing classmates who displayed a consistent interest in interacting with Eve to participate in the study. The peers who participated were all White females between 8 and 9 years old. The first author served as the trainer during all stages of intervention. This study was conducted in accordance with the ethical standards and Institutional Review Board guidelines of the university of the authors. Caregivers consented to child participation prior to baseline.

### 2.2 | Setting and Materials

The study took place in a private Waldorf school. The intervention was conducted in the outdoor recess area and the third-grade classroom. Training, baseline, intervention, booster training, and follow-up sessions were conducted in an outdoor recess area called the Meadow. Generalization probes occurred in the third-grade classroom of 14 students.

A PRT peer training manual containing definitions and pictures illustrating a child using PRT strategies was used. This manual was adapted from training materials created by Pierce and Schreibman (2007). A video camera was used to record all sessions. A basket of Waldorf toys consisting of gnome dolls, wooden animals, tree blocks, and balls was used.

### 2.3 | Response Definitions and Data Collection

All sessions lasted 7 minutes and were video-recorded by the first author. These sessions were analyzed in 15 s intervals, using partial interval recording to provide an estimate of the percentage of intervals in which the behavior occurred (Ledford et al. 2018). The number of intervals in which a target behavior occurred was divided by the total number of intervals and multiplied by 100 to produce an estimated percentage of intervals in which the behavior occurred. Observers could score more than one type of target behavior in the same interval for peers or Eve.

The primary dependent variable was the peers' use of any of the six PRT strategies: (a) orient attention by gaining attention before attempting to give a direction; (b) enhance motivation by providing choices between items and activities; (c) model appropriate and complex play skills by commenting on and narrating play; (d) encourage conversation by getting Eve to ask for desired items and activities; (e) teach taking turns with desired items and activities; and (f) providing praise for engaging in socially appropriate behavior. Definitions of PRT strategies were derived from Pierce and Schreibman's (1995).

In addition to measuring peers' use of PRT, the study included measures of Eve's social communication and play behaviors. Play and communication behaviors were coded using partial interval recording to determine the following: communication to peer, defined as any utterances and vocalizations emitted by Eve directed to a peer, and play with a peer, defined as any time Eve engaged in activities involving the peer.

### 2.4 | Experimental Design and Analysis

A concurrent multiple probe across peers design was used to assess peers' ability to implement PRT and the benefits of intervention for Eve (Gast et al. 2018). All participants began baseline data collection on the same day, and after stable baseline conditions had been established, the first peer began training on PRT. Data on peers' use of PRT and Eve's targeted behaviors were gathered during recess. Intervention sessions continued until the peer reached mastery criteria on their use of PRT and entered follow-up. When the first peer displayed success with PRT use, the following peer was trained on PRT, and the pattern was repeated for all three peers. Visual analysis was used to make conditional change decisions and identify functional relations, as evidenced by peers' increased use of PRT strategies after training or Eve's increased use of communication to peers and play with peers when peers used PRT during recess.

The effect size of the intervention was calculated for peers' use of PRT strategies and Eve's targeted behavior between baseline and intervention, and effect sizes were summarized using Tau-U. Baseline trend adjustments were made whenever the trend was therapeutic so that the adjustment led to a more conservative estimate of the effects size (Parker et al. 2011). Tau-U was calculated using an online calculator (Pustejovsky and Swan 2018). Tau-U is a non-parametric effect size index that



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examines the magnitude of change from baseline to intervention. The accepted metrics for Tau-U are that a result greater than 0.80 represents a very large effect, one between 0.60 and 0.80 a large effect, between 0.20 and 0.60 a moderate effect, and less than 0.20 a small effect (Vannest and Ninci 2015).

## 2.5 | Procedures

### 2.5.1 | Baseline

Prior to training peers on PRT, baseline data were collected during regularly scheduled recess in the Meadow with other children present. During baseline sessions, peers were asked one at a time to play near Eve but were not given any explicit prompts to play with her. Data collection began when the trainer announced, "It's time to play!" and lasted 7 minutes.

### 2.5.2 | Training

Peer training was conducted in the Meadow, with no other students present, for four 30 min training sessions over 2 weeks. The basket of toys was present, and peers were each provided a PRT manual. The first author video-recorded all sessions, acted as the trainer, and taught PRT strategies one at a time. The trainer first defined and explained the strategy (e.g., "Showing good playing means we talk about the toys we are playing with") and then modeled the correct and incorrect use of the strategy with the peer (e.g., giving choices by presenting a ball and wooden tree blocks vs. listing all toys present). The peer was prompted to demonstrate correct and incorrect PRT strategy use during role-play opportunities with the trainer (e.g., "Can you show me a good way to take turns with the gnome doll?"). The trainer provided feedback in the form of specific praise and corrective feedback. The peer was asked questions about PRT strategies to ensure understanding of the concepts (e.g., "What are some ways we can show good playing with the wooden animals?"), and an opportunity was provided to ask questions. During the final training session, the trainer used a role-play checklist to collect data on peers' use of all six PRT strategies during three role-plays and continued until all were completed correctly.

### 2.5.3 | Intervention

Intervention sessions were conducted in the natural environment of the Meadow during recess with other students present. Intervention sessions began when the trainer presented the basket of toys and stated, "It's time to play!". The peer was observed while playing with Eve for 7 minutes.

If a peer did not reach an estimated average PRT use of above 40% of intervals in intervention sessions, a booster training session was provided. This session consisted of an additional 30 min to review and role-play all PRT strategies with trainer feedback. The peer was also provided the opportunity to earn a small reinforcing item for PRT use, and post-session praise and corrective feedback were provided during subsequent intervention sessions.

### 2.5.4 | Generalization

Generalization probes were conducted in the untrained natural environment of the classroom during indoor free play with other students present. During generalization probes, Eve and the peer were not told, "It's time to play!" and the basket of toys was not provided. Generalization sessions lasted for 7 minutes, and no post-session feedback was provided.

### 2.5.5 | Follow-Up

Follow-up occurred after a peer met the criterion of using PRT during at least 50% of intervals for three consecutive intervention sessions. Follow-up sessions were identical to baseline, and peers who received booster training no longer received post-session feedback.

## 2.6 | Inter-Observer Agreement and Procedural Fidelity

Inter-observer agreement (IOA) was calculated for 80% of all sessions using point-by-point agreement (Ledford et al. 2018). After sessions, two observers independently viewed videos of peers interacting with Eve and coded each interval for the occurrence of the target behaviors. Inter-observer agreement was calculated by dividing the number of interval agreements by the number of interval agreements plus interval disagreements and multiplying by 100. IOA for peers' use of any of the PRT strategies was 96.78% (range = 64%–100%). IOA for Eve's communication to peer was 96.38% (range = 82%–100%) and play with peer was 86.33% (range = 61%–100%).

The trainer's adherence fidelity data was collected to assess the degree to which the components of procedures for training peers to use PRT were implemented with integrity. Two observers independently viewed videos of peer training and used a checklist to determine if all training components were used. The trainer's fidelity was 100% across all peer training sessions. IOA was calculated as described above for 25% of training sessions. IOA for measures of training fidelity was 100%.

## 2.7 | Social Validity

Social validity was measured through interviews with peers and a teacher questionnaire. The teacher conducted the peer interviews in the classroom and recorded the peers' responses on a peer interview form developed by the authors for this study based on the peer questionnaire developed by Jones (2007). After the study ended, the teacher completed a questionnaire on her perception of the social interaction changes between Eve and each peer as a result of the intervention and the likelihood of her recommending the intervention to other educators.



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## 3 | Results

### 3.1 | Peers' PRT Use

Figure 1 displays the estimated percentage of intervals that peers implemented any of the six PRT strategies. During baseline, peers used PRT on few occasions, establishing a low level of baseline stability. After training, PRT use increased, displaying immediacy of the effect for two of the three peers and a consistent accelerating trend overall for all three peers. A functional relation between training peers to use PRT and an accelerating trend and increasing PRT strategy use also display inter-participant replication within the study. Almost no overlap between baseline and intervention sessions was observed with all peers. The intervention's effect size was calculated for peers' use of PRT using Tau-U to supplement visual analysis. During

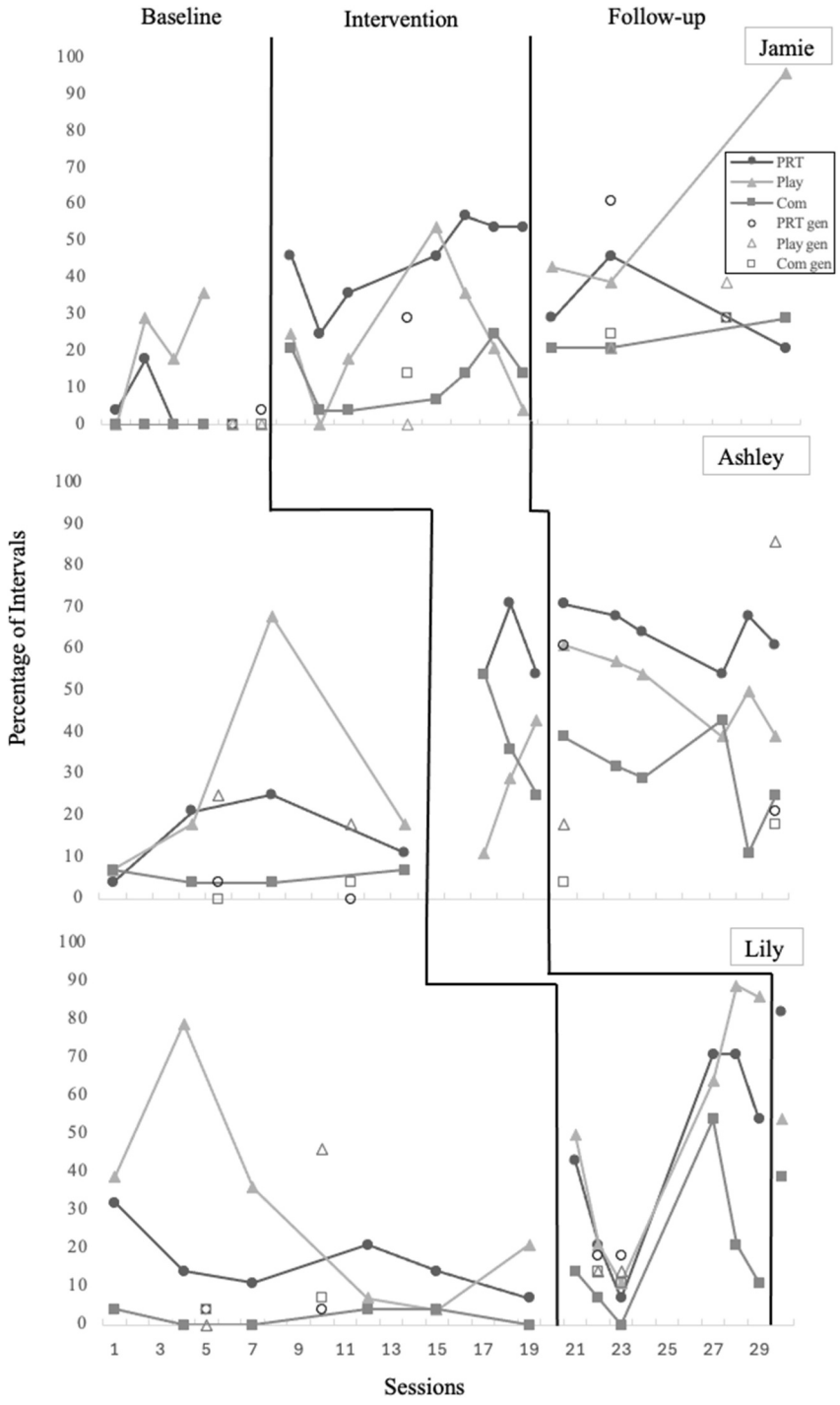



**FIGURE 1** | Peers' PRT use and Eve's targeted behaviors. Solid circles denote the results of each peers' use of all six PRT strategies. Solid triangles represent Eve's use of play with peer (Play). Solid squares indicate Eve's use of communication to peer (Com). Open circles denote the results of each peers' use of all six PRT strategies in generalization probes (PRT gen). The open triangles represent Eve's use of play with peer in generalization probes (Play gen). The open squares indicate Eve's use of communication to peer in generalization probes (Com gen).



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follow-up, one peer's use of PRT was maintained above the criterion level, and another peer's use of PRT decreased but remained above baseline levels. These data provided evidence that peers generalized their use of PRT to the untrained natural environment of classroom free play.

Jamie was the first peer to participate in PRT training. During baseline, her use of PRT occurred at low levels, with an estimated average of 5.5% (range = 0%–18%) of intervals. After the first three intervention sessions, Jamie's use of PRT was above baseline ($M$ = 35.77%, range = 25%–46%) but failed to display the criterion for PRT strategy use, thus qualifying for booster training. Following intervention sessions averaged an estimated 52.75% (range = 46%–57%) of intervals, with 45.43% (range = 25%–57%) across all intervention sessions and during follow-up for 32% (range = 21%–46%). During generalization probes, Jamie's use of PRT increased during intervention ($M$ = 29%) and follow-up ($M$ = 45%). The non-aggregated Tau-U for Jamie's PRT strategy use between baseline and intervention was 1.00, a very large effect.

Ashley was the second peer to participate in training. She used PRT for an estimated average of 15.25% (range = 4%–25%) of intervals during baseline, 59.67% (range = 54%–71%) during intervention, and 64.33% (range = 54%–71%) in follow-up. Ashley used PRT in generalization probes during baseline at low levels; probes were not conducted during intervention because she moved to follow-up quickly and increased in follow-up ($M$ = 41%). The non-aggregated Tau-U for PRT use between baseline and intervention was 0.83, a large effect.

Lily was the final peer to participate in PRT training. During baseline, she used PRT for an estimated average of 16.5% (range = 7%–32%) of intervals. Lily's PRT use increased during intervention sessions initially but decreased ($M$ = 23.67%, range = 7%–43%), so she was provided booster training. Afterward, PRT strategy use averaged an estimated 65.33% (range = 54%–71%) of intervals, with 44.5% (range = 7%–71%) across all intervention sessions. Lily's use of PRT increased to an estimated 82% of intervals in one follow-up session. PRT use in generalization probes increased from baseline to intervention, but no follow-up generalization probes were conducted due to time constraints. The non-aggregated Tau-U for Lily's PRT strategy use between baseline and intervention was 0.61, a large effect.

### 3.2 | Eve's Communication

Figure 1 provides data on Eve's communication behavior with each peer throughout all sessions. During baseline, she engaged in low levels of communication to peers, establishing baseline stability. During intervention and follow-up, she increased her communication to all peers, displaying immediacy of change and a functional relationship to peers' use of PRT.

During baseline with Jaime, Eve engaged in no communication to peer (0% of intervals). During intervention, Eve engaged in communication to peer for an estimated average of 9.67% (range = 4%–21%) of intervals before receiving booster training and 15% (range = 7%–25%) afterward, with 12.71% (range = 4%–25%) overall. In follow-up sessions with Jamie, Eve's communication to peer averaged an estimated 23.67% (range = 21%–29%) of intervals. The non-aggregated Tau-U for Eve's communication to peer with Jamie between baseline and intervention was 1.00, a very large effect. During baseline generalization probes with Jamie, Eve engaged in no communication with peers but increased in intervention and follow-up probes.

Eve engaged in communication to peer for an estimated average of 5.5% (range = 4%–7%) of intervals during baseline with Ashley. In intervention sessions, Eve showed immediacy of change in her communication to peer ($M$ = 38.33%, range = 25%–54%). During follow-up sessions with Ashley, Eve engaged in communication to peer an estimated average of 29.83% (range = 11%–43%) of intervals. The non-aggregated Tau-U for Eve's communication to peer with Ashley between baseline and intervention was 1.00, a very large effect. Generalization probes were conducted in baseline and follow-up. The patterns in her communication were similar across phases, with an increase noted in one of the follow-up generalization probes.

While in baseline with Lily, Eve engaged in communication to peer for an estimated average of 2% (range = 0%–4%) of intervals. Once in intervention, overall, a somewhat variable but accelerating trend was observed in communication ($M$ = 17.83%, range = 0%–54%). Prior to booster training, communication to peer occurred at an estimated average of 7% (range = 0%–14%) of intervals and after 28.67% (range = 11%–54%). During one follow-up session, Eve engaged in communication to peer at an estimated 39% of intervals. The non-aggregated Tau-U for Eve's communication to peer with Lily between baseline and intervention was 0.75, a large effect. During baseline generalization probes with Lily, Eve engaged in very few intervals of communication to peer, but intervention generalization probes showed higher levels.

### 3.3 | Eve's Play

Figure 1 depicts Eve's play behaviors during all sessions. During baseline with all peers, Eve engaged in low levels of play with peers. Once in intervention, a somewhat variable but increasing trend in levels of play with peer was observed with one peer (Lily).

During baseline sessions, Eve primarily engaged in low levels of peer play ($M$ = 20.75%, range = 0%–36%) with Jamie, and throughout intervention sessions, there was an increasing trend ($M$ = 22.57%, range = 0%–54%). Before booster training, play with peer behaviors averaged an estimated 14.33% (range = 0%–25%) of intervals and 28.75% (range = 4%–54%) afterward. Once in follow-up, play with peer was increasing ($M$ = 59.33%, range = 39%–96%). The non-aggregated Tau-U for Eve's play with peer with Jamie between baseline and intervention was 0.04, failing to measure an effect. The pattern for Eve's play with peers in generalization probes with Jamie displayed increases, also observed during follow-up.



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In baseline sessions with Ashley, Eve engaged in play with peer for an estimated average of 22.75% (range = 7%–68%) of intervals, and during intervention, a slight accelerating trend was observed ($M$ = 27.67%, range = 11%–43%). During follow-up sessions, play with peer averaged an estimated 50% (range = 39%–61%) of intervals. The non-aggregated Tau-U for Eve's play with peer with Ashley between baseline and intervention was 0.17, failing to measure an effect. Generalization probes of Eve's play with peer with Ashley conducted in baseline and follow-up only displayed an increase during the final probe.

When in baseline sessions with Lily, Eve engaged in play with peer for an estimated average of 31% (range = 4%–79%) of intervals. Initially, in intervention sessions, play with peer was observed during an estimated average of 27.33% (range = 11%–50%) of intervals, necessitating booster training, and after 73.25% (range = 54%–89%). Overall, a somewhat variable but increasing trend was observed during intervention ($M$ = 53.5%, range = 11%–89%). During the one follow-up session, Eve engaged in play with peer for an estimated 54% of intervals. The non-aggregated Tau-U for Eve's play with peer with Lily between baseline and intervention was 0.47, a moderate effect. Generalization probes of Eve's play with Lily were conducted during baseline and intervention; no change in her play behavior patterns was noted.

### 3.4 | Social Validity

After the study ended, Jamie and Ashley each participated in a brief interview about their participation. Both children responded that they liked using PRT and said it was not too hard to implement. Peers described enjoying learning ways to discover what Eve likes to do. The school year ended before an interview could be conducted with Lily. The teacher completed a questionnaire about peers' use of PRT and agreed that there was an observed improvement in interactions and communication between Eve and peers after intervention.

## 4 | Discussion

The purpose of the study was to examine the effectiveness of training peers in the natural recess environment to deliver PRT implemented as a PMI with Eve, a child with ASD, and the generalization of skills to an untrained, natural, free-play classroom environment. An additional purpose of the study was to examine the effects of peer-implemented PRT on Eve's social communication and play behaviors. This study provides evidence that early elementary school-aged students can learn to use PRT within natural, free-play settings with a child with ASD. Interviews with peers in the study revealed they enjoyed learning and using PRT.

The second research question examined the relationship between peers' use of PRT with Eve and her social communication and play behaviors. There was a functional relationship between the use of PRT and her communication. After peers began using PRT, Eve increased her communication to peers. During sessions when peers used PRT, Eve's play with peers was variable but showed some increases. The study did not demonstrate a functional relation between peers' use of PRT and Eve's play with peers. The classroom teacher also noted the positive effects of the intervention in questionnaire responses after the intervention ended.

The intervention procedures presented provide an example of natural change agents of peers implementing PRT as a PMI to promote social communication and play interactions of a child with ASD in the inclusive classroom setting. The results of this study are consistent with the positive findings of using PRT as a PMI implemented in the early childhood environment to promote the social skills of students with ASD (Pierce and Schreibman 1995, 1997).

While all peers used PRT during generalization probes in the classroom during free play with Eve, Jamie and Ashley only used PRT above criterion level once during follow-up generalization probes. It might be that peers had not reached sufficient fluency in using PRT skills to generalize their use at similar levels or because some of the toys used during intervention sessions were not available in the classroom, and peers may have experienced difficulties generalizing strategies to classroom toys. Future studies should examine the use of strategies (e.g., train common stimuli, sufficient stimulus exemplars) within the training to increase the likelihood of generalization.

### 4.1 | Implications for Future Research and Limitations

Readers should consider several limitations of this study. The first limitation is the failure to demonstrate a functional relationship between peers' use of PRT and Eve's play with peers. It is possible that the dose of the intervention was insufficient to result in changes in Eve's play. While Eve became more communicative with her peers, she did not significantly increase play with them. This might also result from the short duration of the intervention sessions. Future research should examine supporting peers to facilitate PRT with children with ASD in other free-play locations, such as afterschool programs, which provide extended durations of free play time and would allow peers more opportunities to use PRT than peers in the present study.

Another limitation is the lack of generalization measures between peers and other children with ASD, as well as between Eve and untrained peers. It is unknown if peers would have generalized PRT use to play with novel children with ASD or if Eve would have generalized use of communication or play with untrained peers. In their empirical review of peer-mediated interventions for children with ASD, Zagona and Mastergeorge (2018) described the critical need for PMI studies to include generalization measures with novel individuals, activities, and settings.

Although all three peers' use of PRT during follow-up occurred at similar levels observed during intervention, the end of the school year limited the collection of maintenance data for one



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peer and long-term follow-up data for all three peers, presenting a limitation in this study. Chang and Locke (2016) also observed the need for long-term follow-up measures in their systematic review of PMI for children with ASD when they called for exploring intervention effects into the next school year. Long-term evaluation of the effects of PRT should consider evaluating social interactions during the following school year.

## 5 | Conclusion

This study provides data that indicate peer PRT training procedures are effective in changing the behavior of peers toward their classmate with ASD during free play situations in the natural environment. Moreover, these changes in peer interactions resulted in changes in Eve's social communication behaviors. While implementation occurred in only one classroom, the study results suggest that PRT implemented as a PMI might effectively promote peer social interactions within inclusive settings.


### Acknowledgments

This study was conducted in partial fulfillment of the requirements of the first author's master's and doctoral degrees offered through the Department of Behavior and Community Sciences, University of South Florida.


### Ethics Statement

All procedures described herein involving human participants were in accordance with the ethical standards and IRB guidelines of the University of South Florida and with the 1964 Helsinki declaration and its later amendments or comparable standards.

### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

Due to privacy concerns, data that support the findings of this study are available upon reasonable request from the corresponding author. Access to raw data from this study is restricted due to privacy considerations.

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